

CO2 – Assessment Test 5 (AT2)

DSA0603 Data Handling and Visualization | Slot B

Question 1 — Chart Selection (Topic 1: Directory of Visualizations)

For each scenario below, the single best chart type from the directory is identified along with a concise justification.

- 1. Comparing total revenue across 6 product categories in one year.

Selected Chart: — Best Chart Type: Vertical Bar Chart (Amounts)

Justification: — Justification: Best suited for comparing discrete numerical quantities across a small-to-moderate number of categorical groups using visual bar length.

- 2. Showing how the salary distribution differs across 4 job levels.

Selected Chart: — Best Chart Type: Violin Plot or Box Plot (Distribution)

Justification: — Justification: Allows simultaneous comparison of data spread, median, and probability density across multiple distinct groups.

- 3. Showing market share of 5 competing brands.

Selected Chart: — Best Chart Type: Donut Chart or Pie Chart (Proportion)

Justification: — Justification: Effectively displays part-to-whole relationships where components sum to 100% across a small categorical breakdown.

- 4. Showing the relationship between advertising spend and sales revenue.

Selected Chart: — Best Chart Type: Scatter Plot with Trendline (X-Y Relationship)

Justification: — Justification: Enables the visualization of correlation, dispersion, and potential functional dependencies between two continuous numerical variables.

- 5. Showing monthly website traffic over 3 years.

Selected Chart: — Best Chart Type: Line Chart (Temporal)

Justification: — Justification: Ideal for visualizing continuous trends and sequential patterns over ordered time periods.

- 6. Showing which employees report to which managers in an org.

Selected Chart: — Best Chart Type: Network Graph / Tree Diagram (Network / Hierarchical)

Justification: — Justification: Explicitly maps hierarchical parent-child relationships and complex connection structures between entities.

Question 2 — Bar Plot with Highest-Score Highlight

Task: Create a vertical bar plot of Average_Score by Subject, sorted in descending order of score, with a distinct color highlight for the highest-scoring subject and data labels on top of each bar.

Python Implementation

```
import pandas as pd
import matplotlib.pyplot as plt

df_q2 = pd.DataFrame({
    'Subject': ['Mathematics', 'Science', 'English', 'History', 'Computer Sci'],
    'Average_Score': [72, 68, 75, 64, 81]
}).sort_values(by='Average_Score', ascending=False).reset_index(drop=True)

plt.figure(figsize=(8, 5))
colors = ['#d95f02' if score == df_q2['Average_Score'].max() else '#2b5c8f' for score in df_q2['Average_Score']]
bars = plt.bar(df_q2['Subject'], df_q2['Average_Score'], color=colors, width=0.6)

plt.title('Average Score by Subject (Descending Order)', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Subject', fontsize=11, fontweight='bold')
plt.ylabel('Average Score', fontsize=11, fontweight='bold')
plt.ylim(0, 100)

for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, yval + 2, f'{yval}', ha='center', va='bottom', fontweight='bold')

plt.tight_layout()
plt.show()
```

R Implementation

```
library(ggplot2)
library(dplyr)

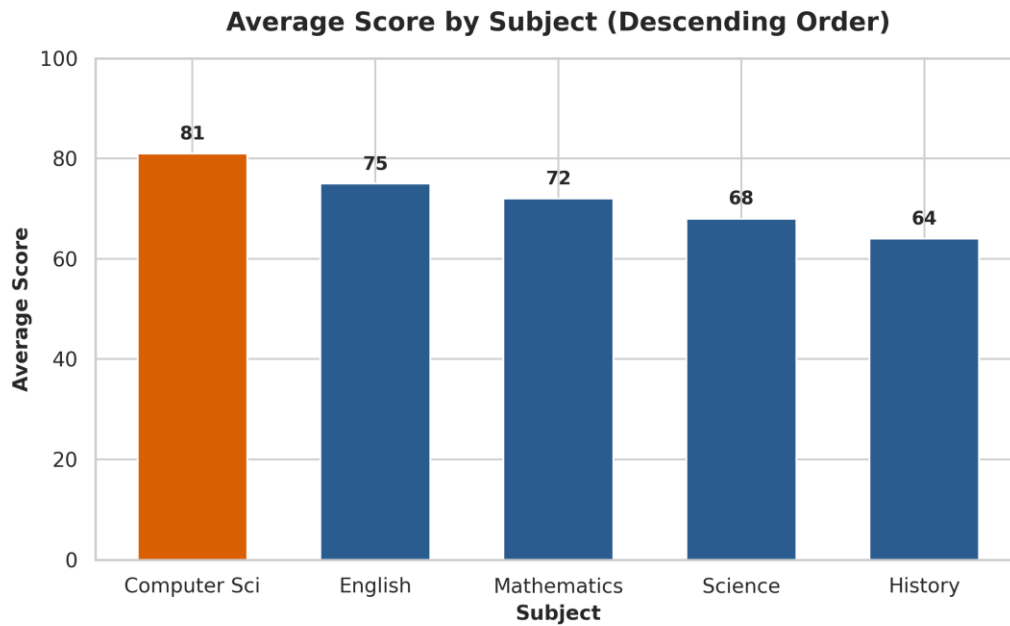
df_q2 <- data.frame(
  Subject = c('Mathematics', 'Science', 'English', 'History', 'Computer Sci'),
  Average_Score = c(72, 68, 75, 64, 81)
) %>% arrange(desc(Average_Score))

df_q2$Subject <- factor(df_q2$Subject, levels = df_q2$Subject)
df_q2$Highlight <- ifelse(df_q2$Average_Score == max(df_q2$Average_Score), 'Highest', 'Other')

ggplot(df_q2, aes(x = Subject, y = Average_Score, fill = Highlight)) +
  geom_col(width = 0.6) +
  scale_fill_manual(values = c('Highest' = '#d95f02', 'Other' = '#2b5c8f')) +
  geom_text(aes(label = Average_Score), vjust = -0.5, fontface = 'bold') +
  ylim(0, 100) +
  labs(title = 'Average Score by Subject (Descending Order)', x = 'Subject', y = 'Average Score') +
```

```
theme_minimal() +  
theme(legend.position = 'none', plot.title = element_text(face = 'bold', size = 14))
```

Visualization Output & Interpretation



Interpretation: Computer Science leads significantly with an average score of 81, highlighted in distinct orange. English (75) and Mathematics (72) follow closely behind, while History records the lowest average score at 64. Sorting in descending order allows immediate ranking comparison.

Question 3 — Multi-View Regional Sales Analysis

Task: Produce four distinct visual views from the regional sales dataset: (1) Grouped bar chart, (2) Stacked bar chart, (3) Ranked dot plot of annual sales, and (4) Sales heatmap.

Python Implementation (Reshaping & Plotting)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data_q3 = [
    ['North', 'Q1', 120], ['North', 'Q2', 135], ['North', 'Q3', 128], ['North', 'Q4', 150],
    ['South', 'Q1', 95], ['South', 'Q2', 110], ['South', 'Q3', 105], ['South', 'Q4', 130],
    ['East', 'Q1', 140], ['East', 'Q2', 138], ['East', 'Q3', 145], ['East', 'Q4', 160],
    ['West', 'Q1', 80], ['West', 'Q2', 85], ['West', 'Q3', 90], ['West', 'Q4', 100]
]
df_q3 = pd.DataFrame(data_q3, columns=['Region', 'Quarter', 'Sales_000s'])
pivot_grouped = df_q3.pivot(index='Region', columns='Quarter', values='Sales_000s')[['Q1',
'Q2', 'Q3', 'Q4']]

# 1. Grouped Bar
pivot_grouped.plot(kind='bar', figsize=(8, 4.5), colormap='viridis', width=0.8)
plt.title('Sales by Region and Quarter (Grouped)', fontweight='bold')
plt.ylabel('Sales (000s)'); plt.xticks(rotation=0); plt.show()

# 2. Stacked Bar
pivot_grouped.plot(kind='bar', stacked=True, figsize=(8, 4.5), colormap='magma', width=0.6)
plt.title('Sales by Region and Quarter (Stacked)', fontweight='bold')
plt.ylabel('Total Sales (000s)'); plt.xticks(rotation=0); plt.show()

# 3. Dot Plot
annual_sales =
df_q3.groupby('Region')['Sales_000s'].sum().reset_index().sort_values('Sales_000s')
plt.figure(figsize=(7, 4))
plt.scatter(annual_sales['Sales_000s'], annual_sales['Region'], color='#1f77b4', s=120)
plt.title('Total Annual Sales by Region', fontweight='bold'); plt.show()

# 4. Heatmap
plt.figure(figsize=(7, 4))
sns.heatmap(pivot_grouped, annot=True, fmt='d', cmap='Blues')
plt.title('Sales Heatmap by Region & Quarter', fontweight='bold'); plt.show()
```

R Implementation

```
library(tidyverse)

df_q3 <- data.frame(
  Region = rep(c('North', 'South', 'East', 'West'), each = 4),
  Quarter = rep(c('Q1', 'Q2', 'Q3', 'Q4'), times = 4),
  Sales_000s = c(120, 135, 128, 150, 95, 110, 105, 130, 140, 138, 145, 160, 80, 85, 90,
100)
)

# 1. Grouped Bar
```

```

ggplot(df_q3, aes(x = Region, y = Sales_000s, fill = Quarter)) +
  geom_col(position = 'dodge') + labs(title = 'Grouped Bar Chart')

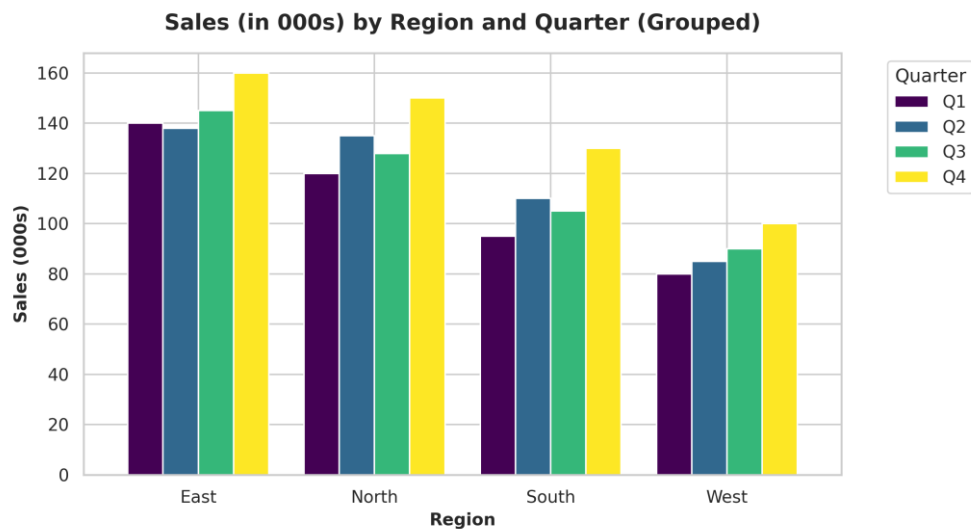
# 2. Stacked Bar
ggplot(df_q3, aes(x = Region, y = Sales_000s, fill = Quarter)) +
  geom_col(position = 'stack') + labs(title = 'Stacked Bar Chart')

# 3. Dot Plot
annual_sales <- df_q3 %>% group_by(Region) %>% summarise(Total = sum(Sales_000s))
ggplot(annual_sales, aes(x = Total, y = reorder(Region, Total))) +
  geom_point(size = 4, color = '#1f77b4') + labs(title = 'Ranked Dot Plot')

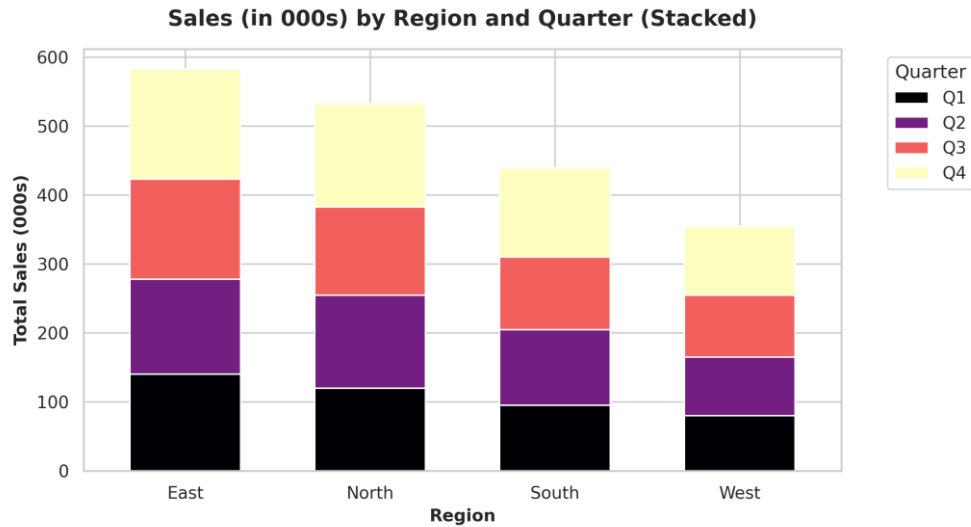
# 4. Heatmap
ggplot(df_q3, aes(x = Quarter, y = Region, fill = Sales_000s)) +
  geom_tile() + scale_fill_gradient(low = '#deebf7', high = '#08519c')

```

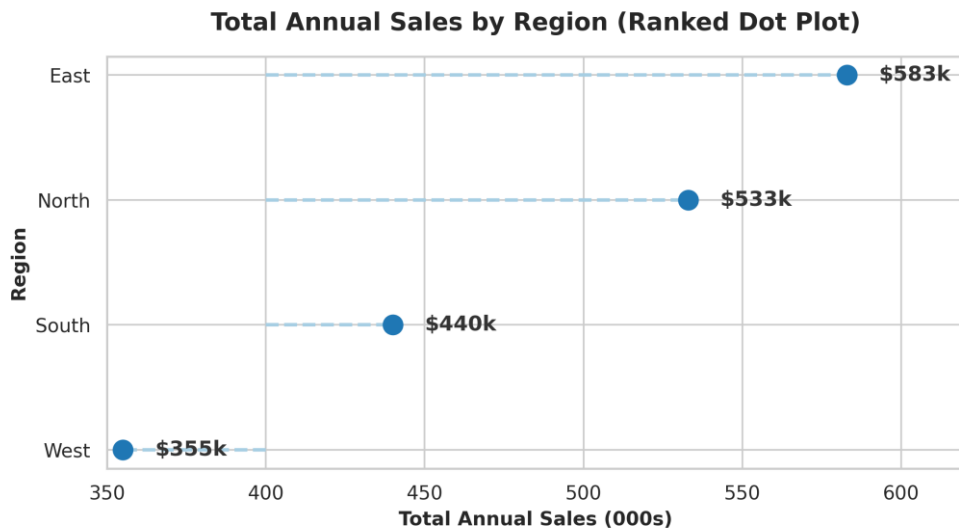
Visualization Outputs & Interpretations



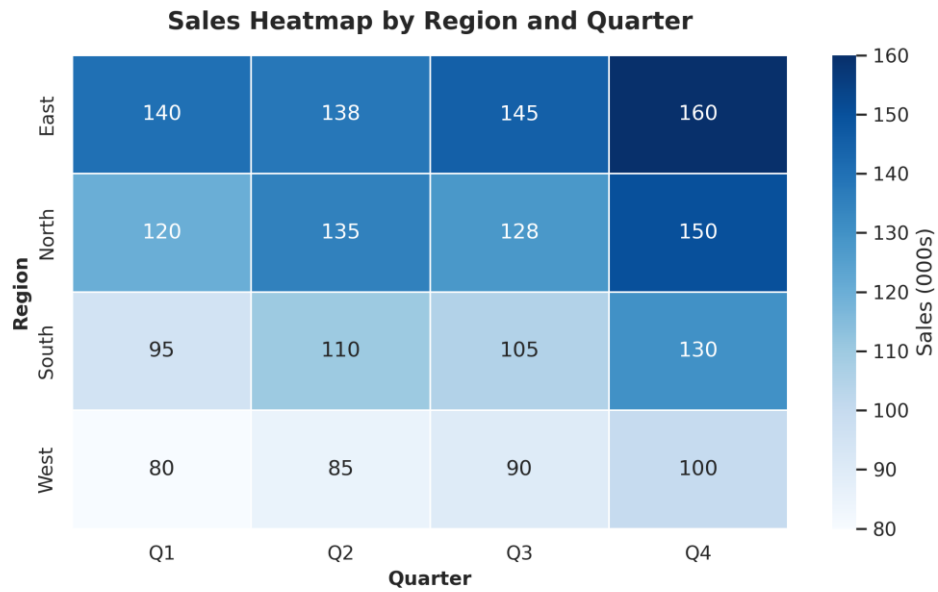
Interpretation (Grouped Bar): East consistently outperforms other regions across Q1 to Q4, reaching a peak of \$160k in Q4. West records the lowest sales across all quarters.



Interpretation (Stacked Bar): Clearly illustrates cumulative annual totals per region while showing seasonal contribution breakdown. East leads with ~\$583k total, followed by North (\$533k), South (\$440k), and West (\$355k).



Interpretation (Ranked Dot Plot): Provides a clean, uncluttered ranking of total annual sales by region, highlighting East as the top performer and West at the baseline.



Interpretation (Heatmap): The color intensity matrix immediately draws attention to East in Q4 (darkest blue, 160) as the highest single quarter-region combination, and West in Q1 (lightest blue, 80) as the lowest.

Question 4 — Synthetic Class Exam Scores (Violin & Ridgeline Plots)

Task: Generate synthetic exam scores for 4 classes with random seed 42 (Class A: $N(70,8)$, Class B: $N(75,5)$, Class C: $N(65,12)$, Class D: $N(80,6)$) and produce a Violin plot and a Ridgeline plot ordered by mean score.

Python Implementation

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(42)
scores_a = np.random.normal(70, 8, 40)
scores_b = np.random.normal(75, 5, 40)
scores_c = np.random.normal(65, 12, 40)
scores_d = np.random.normal(80, 6, 40)

df_q4 = pd.DataFrame({
    'Score': np.concatenate([scores_a, scores_b, scores_c, scores_d]),
    'Class': ['Class A']*40 + ['Class B']*40 + ['Class C']*40 + ['Class D']*40
})

# Violin Plot
plt.figure(figsize=(8, 4.5))
sns.violinplot(x='Class', y='Score', data=df_q4, palette='Set2', inner='box', cut=0)
plt.title('Exam Score Distribution by Class (Violin Plot)', fontweight='bold')
plt.show()

# Ridgeline Plot (Ordered by mean: C, A, B, D)
class_order = ['Class C', 'Class A', 'Class B', 'Class D']
palette = sns.color_palette("husl", 4)
fig, axes = plt.subplots(4, 1, figsize=(8, 5), sharex=True)
for i, cls in enumerate(class_order):
    subset = df_q4[df_q4['Class'] == cls]['Score']
    sns.kdeplot(subset, ax=axes[i], fill=True, color=palette[i], alpha=0.6)
    axes[i].set_ylabel(cls, rotation=0, labelpad=40, fontweight='bold')
    axes[i].set_yticks([]); axes[i].grid(False)
plt.suptitle('Ridgeline Plot of Exam Scores by Class', fontweight='bold', y=0.98)
plt.show()
```

R Implementation

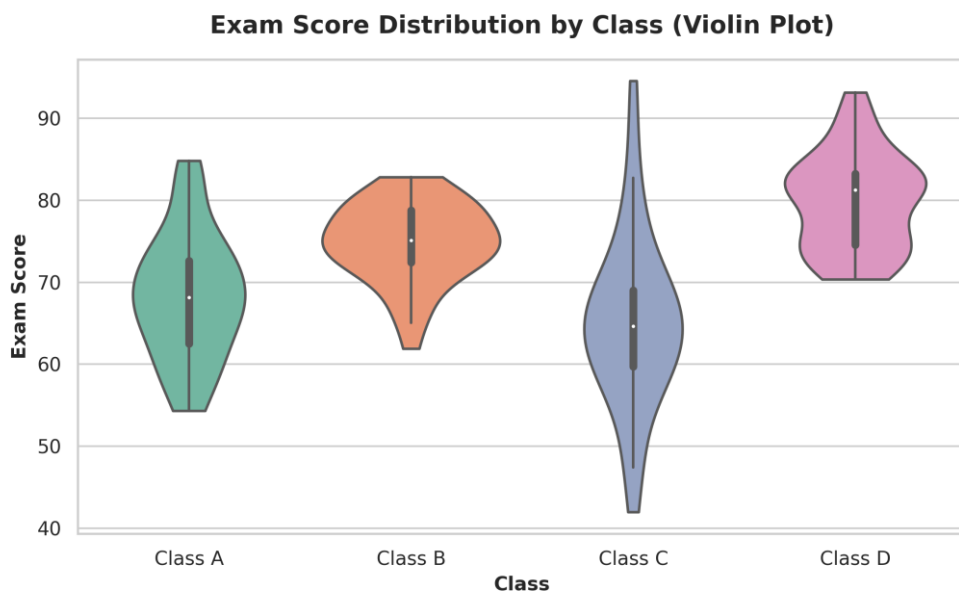
```
library(tidyverse)
library(ggribes)

set.seed(42)
df_q4 <- data.frame(
  Score = c(rnorm(40, 70, 8), rnorm(40, 75, 5), rnorm(40, 65, 12), rnorm(40, 80, 6)),
  Class = rep(c('Class A', 'Class B', 'Class C', 'Class D'), each = 40)
)
```

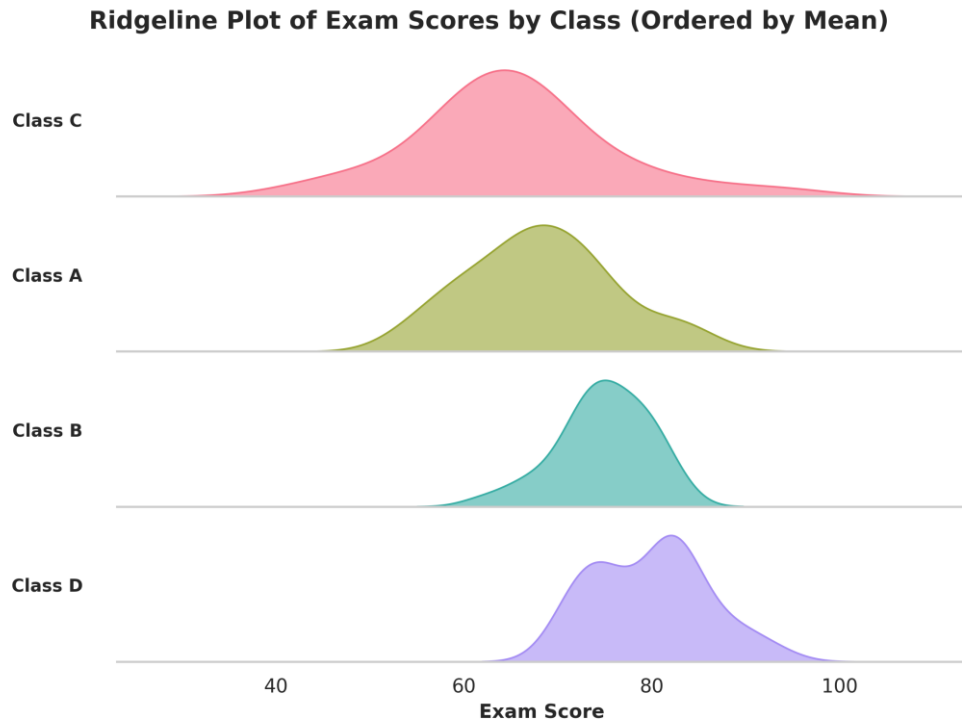
```
# Violin Plot
ggplot(df_q4, aes(x = Class, y = Score, fill = Class)) +
  geom_violin(trim = FALSE) + geom_boxplot(width = 0.1, color = 'black', alpha = 0.7) +
  labs(title = 'Exam Score Distribution by Class (Violin Plot)')

# Ridgeline Plot
df_q4$Class <- factor(df_q4$Class, levels = c('Class D', 'Class B', 'Class A', 'Class C'))
ggplot(df_q4, aes(x = Score, y = Class, fill = Class)) +
  geom_density_ridges(alpha = 0.7, scale = 1.5) +
  labs(title = 'Ridgeline Plot of Exam Scores by Class')
```

Visualization Outputs & Interpretations



Interpretation (Violin Plot): Class D exhibits the highest median score (~80) with a compact spread (SD=6). Class C shows the widest dispersion and lowest median (~65) with heavy tails ranging from 42 to 95. Class B is tightly clustered around 75 with low variance (SD=5).



Interpretation (Ridgeline Plot): Ordered sequentially by mean score (Class C -> A -> B -> D), the ridgeline plot effectively highlights the progressive rightward shift in central tendency and the distinct differences in distributional width and skewness across the four classes.

Question 5 — Distributional Shape & KDE Analysis of Tips Dataset

Task: Plot a histogram of the continuous variable `total_bill` (30 bins) with an overlaid KDE curve, and repeat split by the smoker categorical variable to compare distribution shapes.

Python Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset('tips')

# 1 & 2. Overall Histogram with KDE
plt.figure(figsize=(8, 4.5))
sns.histplot(tips['total_bill'], bins=30, kde=True, color='#2b5c8f', stat='density')
plt.title('Distribution of Total Bill with KDE Overlay', fontweight='bold')
plt.xlabel('Total Bill ($)'); plt.ylabel('Density'); plt.show()

# 3. Grouped by Smoker
plt.figure(figsize=(8, 4.5))
sns.histplot(data=tips, x='total_bill', hue='smoker', bins=30, kde=True, element='step',
stat='density', common_norm=False, palette='Set1')
plt.title('Total Bill Distribution and KDE Split by Smoker Status', fontweight='bold')
plt.xlabel('Total Bill ($)'); plt.ylabel('Density'); plt.show()
```

R Implementation

```
library(ggplot2)
library(reshape2)

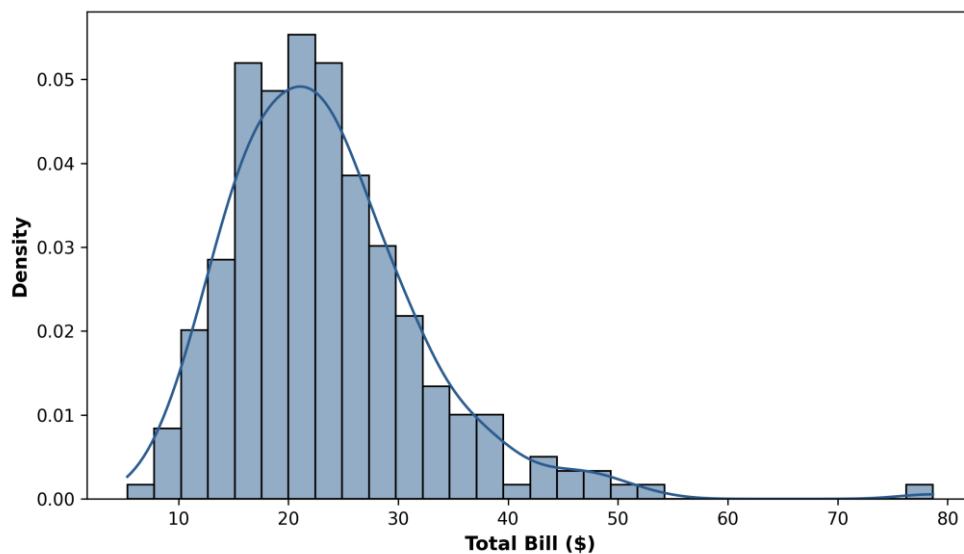
data(tips)

# 1 & 2. Overall Histogram with KDE
ggplot(tips, aes(x = total_bill)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = '#2b5c8f', color =
'white', alpha = 0.7) +
  geom_density(color = '#08519c', linewidth = 1) +
  labs(title = 'Distribution of Total Bill with KDE Overlay', x = 'Total Bill ($)', y =
'Density') +
  theme_minimal()

# 3. Grouped by Smoker
ggplot(tips, aes(x = total_bill, fill = smoker, color = smoker)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, alpha = 0.4, position =
'identity') +
  geom_density(aes(y = after_stat(density)), linewidth = 1, alpha = 0.2) +
  labs(title = 'Total Bill Distribution Split by Smoker Status', x = 'Total Bill ($)', y =
'Density') +
  theme_minimal()
```

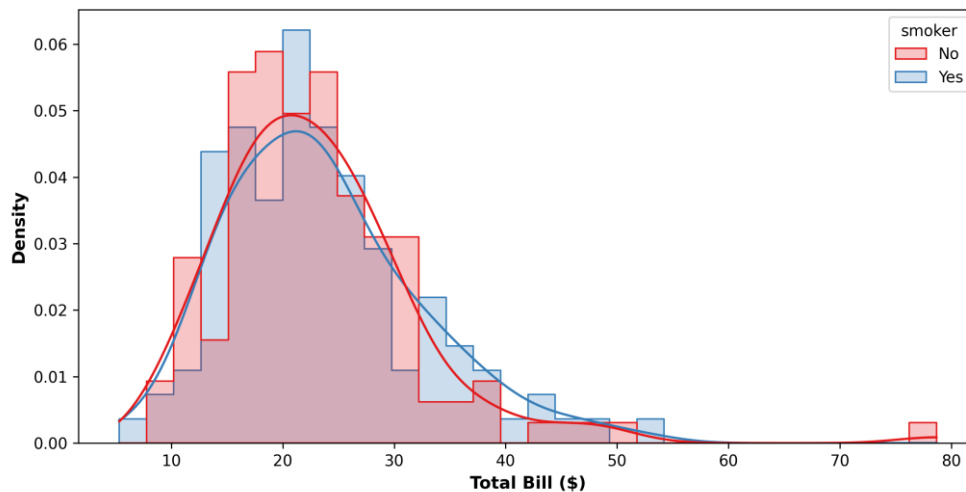
Visualization Outputs & Interpretations

Distribution of Total Bill with KDE Overlay



Interpretation (Overall Histogram & KDE): The total bill distribution is right-skewed, with the majority of bills concentrated between \$10 and \$30, and a long right tail extending up to \$50+. The overlaid KDE curve captures this unimodal, positively skewed density profile smoothly.

Total Bill Distribution and KDE Split by Smoker Status



Interpretation (Grouped by Smoker): Both smoker and non-smoker distributions exhibit right-skewness. However, non-smokers (red) display a slightly higher peak concentration around \$15-\$20, whereas smokers (blue) show a marginally broader spread into higher bill amounts (above \$40).